

**MORTALITY RATE PREDICTION AND ANALYSIS USING
MACHINE LEARNING DUE TO LEADING DISEASES**

Abstract

One of the most common uses of machine learning algorithms has been in the healthcare industry. One of these applications is the identification and prediction of mortality rates. The study's goal is to create a software model that predicts death rates using machine learning. The information, which includes of mortality data from various causes, was collected via the government website [catalog.gov](https://data.cdc.gov). The death rate was predicted using the linear regression technique. The created model functioned well, with a high accuracy rate in forecasting the morality date owing to various reasons. It was discovered in recent years. Covid-19 has been identified as one of the leading causes of human death, which is consistent with real-world observations.

Acknowledgments

I am writing to express my heartfelt gratitude towards our university and professor for giving me the opportunity to conduct the research process. I would like to thank my parents and friends for their constant support.

Table of Contents

Chapter 1: Introduction.....	5
1.1 Introduction.....	5
1.2 Background.....	5
1.3 Problem Statement.....	5
1.4 Research aim.....	6
1.5 Research Objectives.....	6
1.6 Research Questions.....	6
1.7 Research Rationale.....	7
1.8 Report Overview.....	8
1.9 Summary.....	9
Chapter 2: Literature Review.....	10
2.1 Introduction.....	10
2.2 ML Prediction Algorithm for Mortality Pattern Understanding.....	10
2.3 Assessing Features Contributing to the Mortality of the Population.....	13
2.4 Understanding the Importance of Mortality Pattern Prediction.....	15
2.5 Application of ML Algorithms on Mortality Pattern Understanding.....	17
2.6 Theories and Models.....	18
2.7 Literature Gap.....	20
2.8 Summary.....	20
Chapter 3 Methodology.....	20
3.1 Introduction.....	20
3.2 Data Collection Methods.....	21
3.3 Data Preparation.....	22
3.4 Data Analysis Techniques.....	23
3.5 Model Development.....	23

3.6 Project Management Methodology.....	25
3.7 Summary.....	25
Chapter 4: Requirements.....	25
4.1 Introduction.....	25
4.2 Functional Requirements.....	26
4.3 Non-Functional Requirements.....	26
4.4 Tools and Techniques Used.....	27
4.5 Summary.....	27
Chapter 5: Analysis.....	28
5.1 Introduction.....	28
5.2 Logistic Regression.....	28
5.3 Implementation.....	30
5.4 Summary.....	31
Chapter 6: Design.....	32
6.1 Introduction.....	32
6.2 Dataset Modeling.....	32
6.3 Summary.....	33
Chapter 7: Implementation.....	33
7.1 Introduction.....	33
7.2 Process.....	34
7.3 Summary.....	37
Chapter 8: Testing.....	38
8.1 Introduction.....	38
8.2 Process.....	38
8.3 Summary.....	42

Chapter 9: Project Management.....	42
9.1 Schedule Development.....	42
9.2 Risk Management.....	44
7.3 Quality Management.....	44
7.4 Professional, ethical, social, and legal Considerations.....	44
Chapter 10: Critical Appraisal.....	44
Chapter 11: Conclusion.....	44
11.1 Conclusion.....	44
11.2 Achievements.....	44
11.3 Future Work.....	44
Chapter 12: Reflection.....	44
Reference List.....	44

Chapter 1: Introduction

1.1 Introduction

In public health services, understanding the rate of mortality is one of the most important in healthcare. ML algorithms and predictive analytics have popularly been used to realise mortality patterns due to different reasons (Bray, Gibson and White, 2020). This has spread to understand mortality patterns in other causes. For example, during the COVID-19 pandemic, researchers used predictive analytics to understand the mortality rate due to covid-19. In this research process, an overall view of mortality rate prediction due to different diseases and causes will be assessed. This introductory section mainly provides the blueprint of the conducted research by providing the research purpose, background, and overall structure of the report.

1.2 Background

Mortality means mainly the rate of death within a population. This is considered as an important metric to understand the severity of any disease or cause. Due to this, it has been considered a topic of research for a long time. However, for a long time, it was just the fundamental analysis of statistics and studying the different demography. In the 17th century, John Graunt analysed the birth and death records in London (Harkness, 2020). This was the first instance of demographic analysis and mortality rate understanding ever. Since then, understanding the mortality patterns due to different diseases has been a topic of research for a long time. For example, John Snow researched the cholera epidemic in 1854 and made significant progress in the mortality rate understanding (Walford, 2020). During the 1950s, lung cancer-related research was also related to the mortality rate understanding.

The advancement of various types of statistical tools and modern technology has changed the understanding of mortality rate patterns. Machine learning as well as the predictive analytics have provided new avenues in this regard. This was helpful in comparing mortality rates in various populations and finding new ways of diagnosis.

1.3 Problem Statement

There are different factors which are directly linked with mortality patterns. This includes demography, economic and social status, healthcare quality, and various diseases (Alghatani *et al.*, 2021). It is tough to understand the future trends assessing all of these factors. The factors are also different in different regions, and this is a major problem. These are the main problems of this research process, which will be assessed in this research process. Understanding the feature among

them, which is most important for a particular region, is also a complex task. It is tough with traditional techniques to analyse the feature. Due to this reason, ML algorithms would significantly be important to understand the mortality patterns based on the collected dataset.

1.4 Research aim

The research has the purpose of understanding the pattern of mortality rate using ML algorithms.

1.5 Research Objectives

1. To collect and prepare a dataset using which mortality patterns can be understood.
2. To identify the best features which are contributing to the mortality of the population.
3. To develop the software model using ML algorithms for mortality rate prediction.
4. To perform different evaluation techniques to understand the success of this software development.

1.6 Research Questions

1. What type of dataset does mortality rate prediction research need?
2. What are the best features which are contributing to the mortality of the population?
3. What is the best ML algorithm to develop mortality rate prediction software?
4. What are the different evaluation techniques to understand the success of this software development?

1.7 Research Rationale

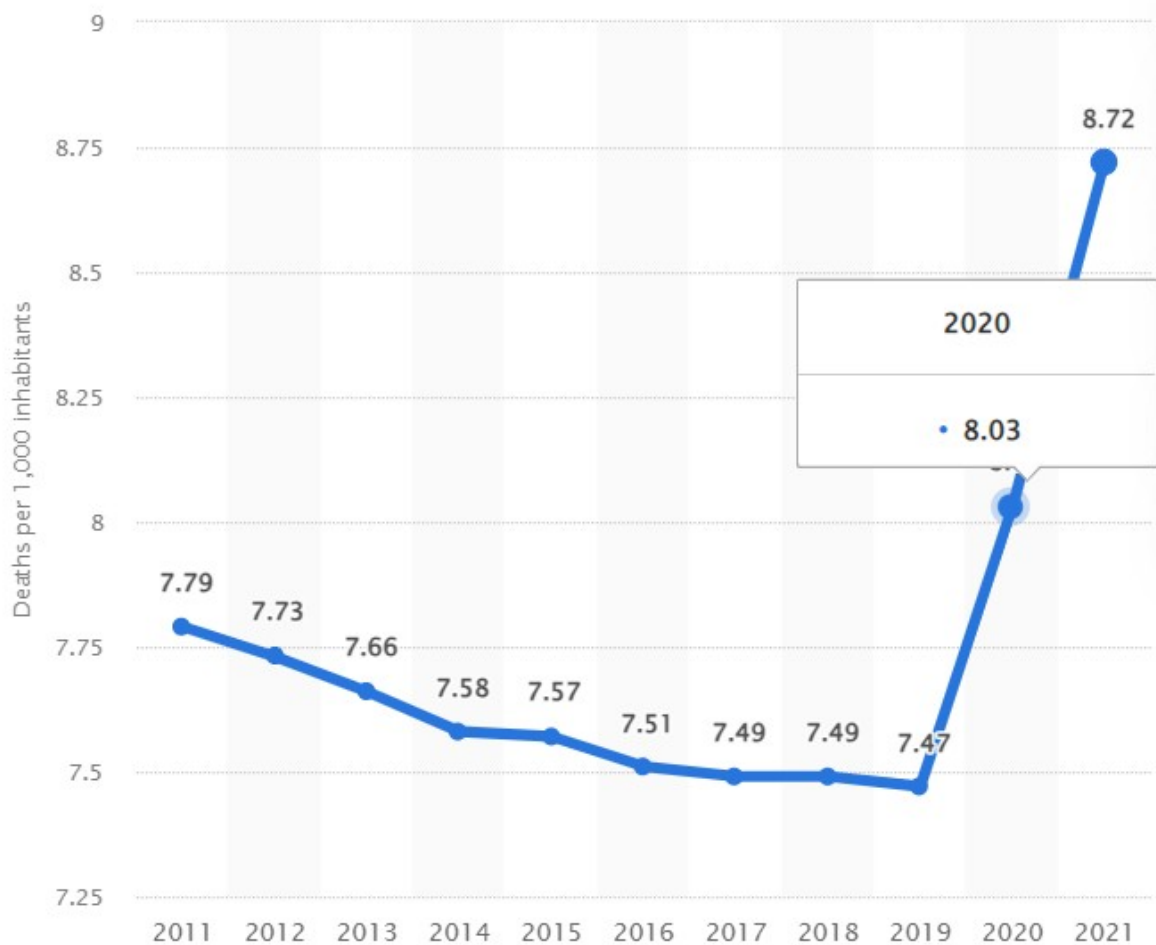


Figure 1.1: Mortality rate worldwide

(Source: Statista, 2022)

The figure shows the worldwide mortality rate from 2011 to 2021. As shown in the figure, the mortality rate significantly increased after 2019 due to the pandemic. It was tough to predict a pandemic like COVID-19 beforehand (Statista, 2022). However, if accurate prediction of mortality rate can be done, then prevention of the diseases could get easier. This is the main rationale and motivation for conducting this research process. The primary reason is to understand the overall well-being of the population. As though mortality pattern identification policies, interventions, and allocation of resources would get easier. One of the main reasons behind conducting the research process was to be able to make a timely intervention to save people's lives. This research could be

crucial for healthcare and the government to save people from different diseases. This could also be helpful to understand or assess similar pandemics in the future beforehand.

Modern data analytics have popularly been used in healthcare, and mortality pattern understanding could be one of the many applications. This research process could have a long-term impact on people as personalised medicine can be provided to patients based on the results of this research process.

Finally, this research problem is evolving and future academic scholars and professionals may research on this topic. The motivation was to help future academic scholars undertake this problem as research and provide them with the required background knowledge on this topic.

1.8 Report Overview

The overview of the research report is as follows:

Introduction: The research report has started with this section, and here, the general overview of this research was provided. This includes the research problem, background, and purpose of this study.

Literature Review: This section will provide in-depth background knowledge on this topic. This includes studying and reviewing existing papers on the topic and discussing theories and models of the research process.

Methodology: The methodological approach of this research process will be provided in this chapter. This includes the process of data collection, analysis, and model development.

Requirements: The main requirements of the research, including the functional and non-functional requirements, will be explained in this chapter.

Analysis: Based on the developed model, in this section, the analysis will be made based on images.

Design: The design process of the software will be discussed in this part with images and justification.

Implementation: The implementation procedure of mortality rate prediction will be discussed in this chapter.

Testing: The model evaluation will be done in this chapter based on the required metrics. The target will be to show the accuracy of the model in this chapter.

Project Management: The project management, including risk management, schedule, and legal and ethical considerations, will be explained in this chapter.

Critical Appraisal: The critical reflection of this research process will be provided using the required tools.

Conclusions: The research will be concluded in this chapter. The recommendations and future scope of this research will be explained in this chapter as concluding remarks.

1.9 Summary

To summaries, mortality pattern understanding is the process of predicting mortality patterns based on different factors. This research process has used ML algorithms to develop a model which would predict mortality patterns. The various aspects and different opinions are the main problems for this research process. This research will be significantly helpful in understanding the interventions, policies, and allocation of resources in healthcare. This research process has great societal value, and it has great importance to future academic scholars as well. The structure of the research process has been provided in this section and an explanation of each part has been effectively made.

Chapter 2: Literature Review

2.1 Introduction

The mortality pattern understanding using predictive modelling has significant societal value, and it requires accurate results to apply it in real life. Due to the importance of this topic, there are many research processes which have been made on this problem. Some have specifically focused on a particular disease like COVID-19 or cancer. On the other hand, some have focused on overall mortality based on a specific population. In this chapter, studying all of these different chapters will be done based on developed themes. The related theories and models will be discussed in this chapter. Finally, the target will be to find limitations of the existing research process so that those can be mitigated in this research process.

2.2 ML Prediction Algorithm for Mortality Pattern Understanding

As per Subudhiet *al.* (2021), ML algorithms can significantly be useful to assist healthcare doctors with individuals at high risk due to COVID-19. There are 18 ML algorithms used by the researchers, and their performance was compared to understand the best algorithm for mortality rate prediction. The authors collected a dataset from March to July of 2020 who were admitted to the emergency ward.

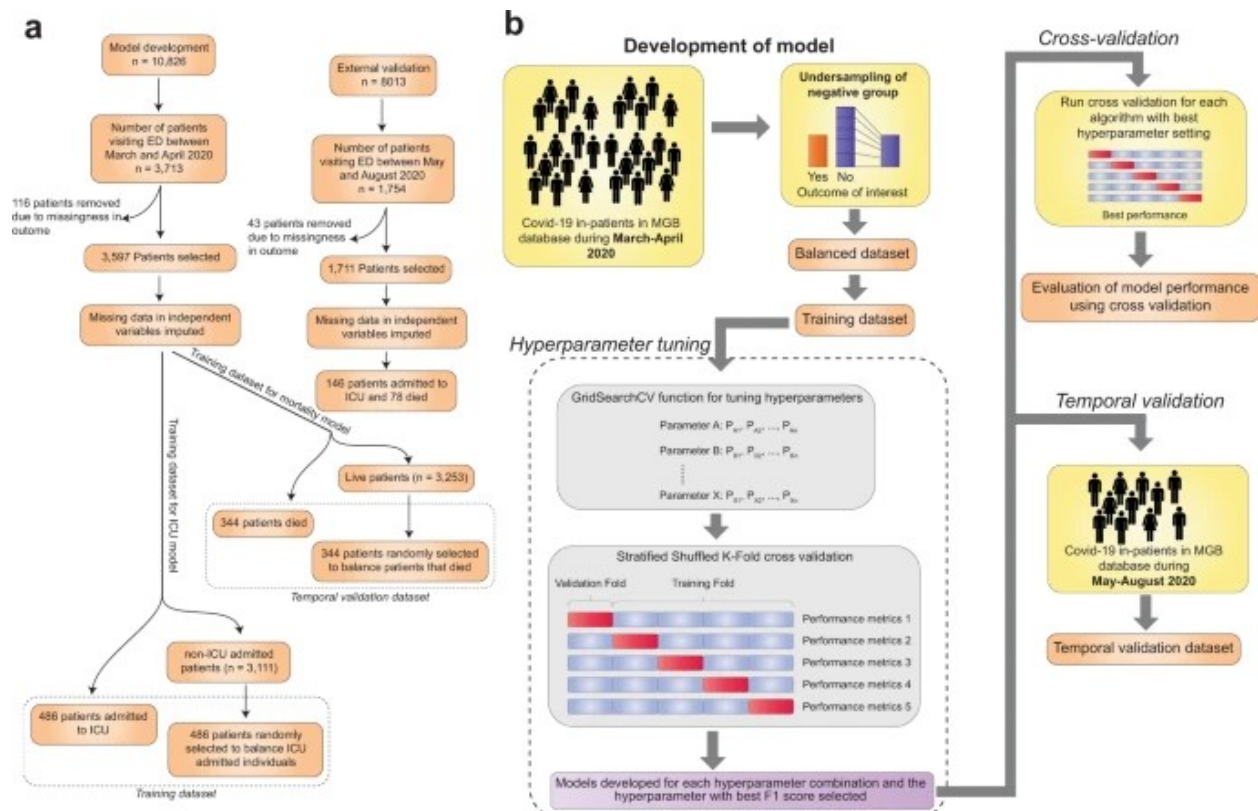


Figure 2.1: Model Development

(Source: Subudhiet *al.*, 2021)

The figure shows how the authors have developed the mortality models using different ML approaches. First, the dataset of patients was collected, and then they were divided for training and testing for model training and evaluation. The statistics of patient death and ICU admission were also included.

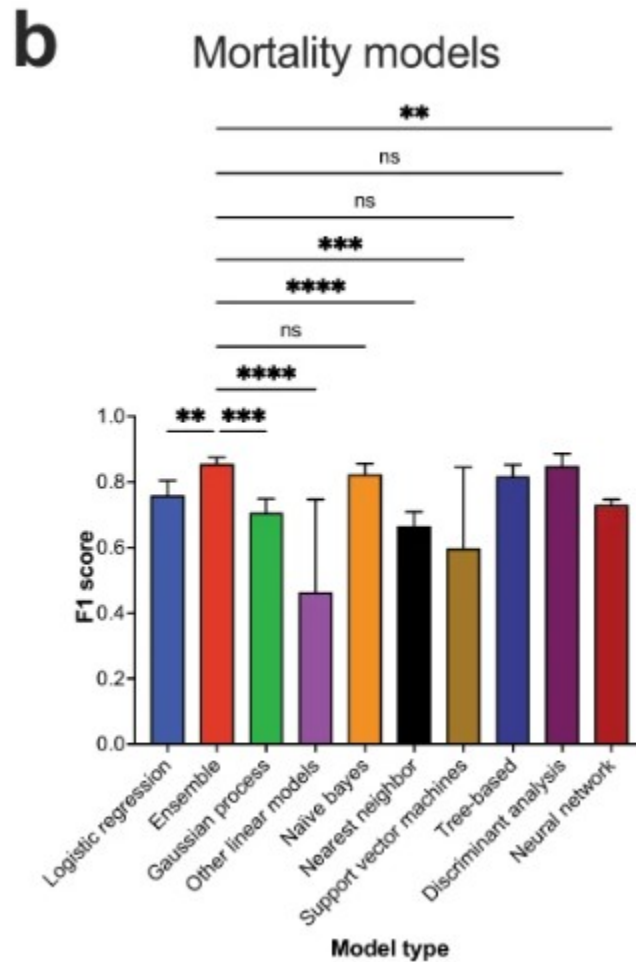


Figure 2.2: Model Comparison

(Source: Subudhiet *al.*, 2021)

The figure shows the performance of different models in mortality rate prediction. As per the figure, all the ensemble model shows the highest F1 score. The authors have also understood the importance of the Random Forest algorithm and developed separate maps for that. Naive Bayes

and discriminant analysis have also shown higher accuracy for mortality rate prediction. On the other hand, KNN and SVM have shown the lowest accuracy while predicting the mortality rate. As per Manzet *et al.* (2020), ML algorithms have the ability to be useful for cancer patients with short mortality to provide treatment planning. However, in oncology ML algorithms have not been particularly used to understand the mortality rate. A dataset of oncology patients from March to April 2019 was used to train the model, and then it predicted the mortality rate for the next 180 days based on the data. It indicates the rolls of mortality between 4 to 8 days and helps the doctors to develop the risk treatment plan. The researchers used a “Gradient-boosting ML binary classifier” to develop the software model. After developing the model, it was integrated with the electronic health record to provide real-time prediction of the mortality of patients within 4 to 8 days. The researchers found that it has outperformed the traditional methods and prognosis used in the hospital. This signifies ML algorithms can indeed improve the diagnosis of cancer patients based on understanding the mortality pattern of the short-time patients.

According to Shin *et al.* (2020), ML algorithms would provide better results compared to the traditional method while predicting mortality. The researchers have compared the mortality patterns of heart failure for ML methods and traditional methods. The researchers here have used secondary data where they have collected journals from the last 20 years related to the problem and discussed their findings. Researchers have evaluated that logistic regression models and other tree-based models have provided the best results and they have certainly outperformed the traditional methods.

On the other hand, another Covid-19 mortality-related research was made by Vaidet *et al.*, (2020). As per Vaid et al., (2020), mortality patterns of Covid-19 patients can effectively be understood using ML models. It is true that the ML model requires a large dataset. However, the authors were able to predict the mortality of COVID-19 patients within 7 days. They have used a logistic regression algorithm with lasso and “multilayer perceptron”. The authors found that their model has outperformed the local model by quite a margin. This way the federated learning is showing promise to be used in hospitals for mortality rate prediction of different diseases.

As per Khan et al. (2020), it is essential that mortality prediction is done based on the current diseases rather than only depending on historical data. This is why the researchers have used COVID-19 pandemic data to understand the mortality rate. The researchers have used regression techniques and then optimised that technique by hyperparameter tuning. They have collected

COVID-19 data from five countries. Those are Pakistan, Turkey, Spain, Sweden, and France and using an ML algorithm the mortality rate was predicted. The developed “GPR-based mortality model” showed that high rate of mortality in Pakistan. It has also shown that the low RMSE score for the country. It was found that the Sweden model developed for that country was the best model to tackle the mortality rate. This model could significantly be helpful for understanding similar diseases in the future as well.

2.3 Assessing Features Contributing to the Mortality of the Population

As per Grekousiset *al.* (2022), there are different factors other than the disease itself that need to be considered while understanding the mortality rate. This was true for COVID-19 as various socio-economic factors have contributed to death due to the pandemic. This was understood due to the disparities in the spatial distribution. This is why the authors have developed a model based on Radom Forest and OLS with the geographically distributed model. As per the study, the lack of health insurance was the most important factor which was evaluated based on the US data. Lack of physical activity was also evaluated to be the reason here.

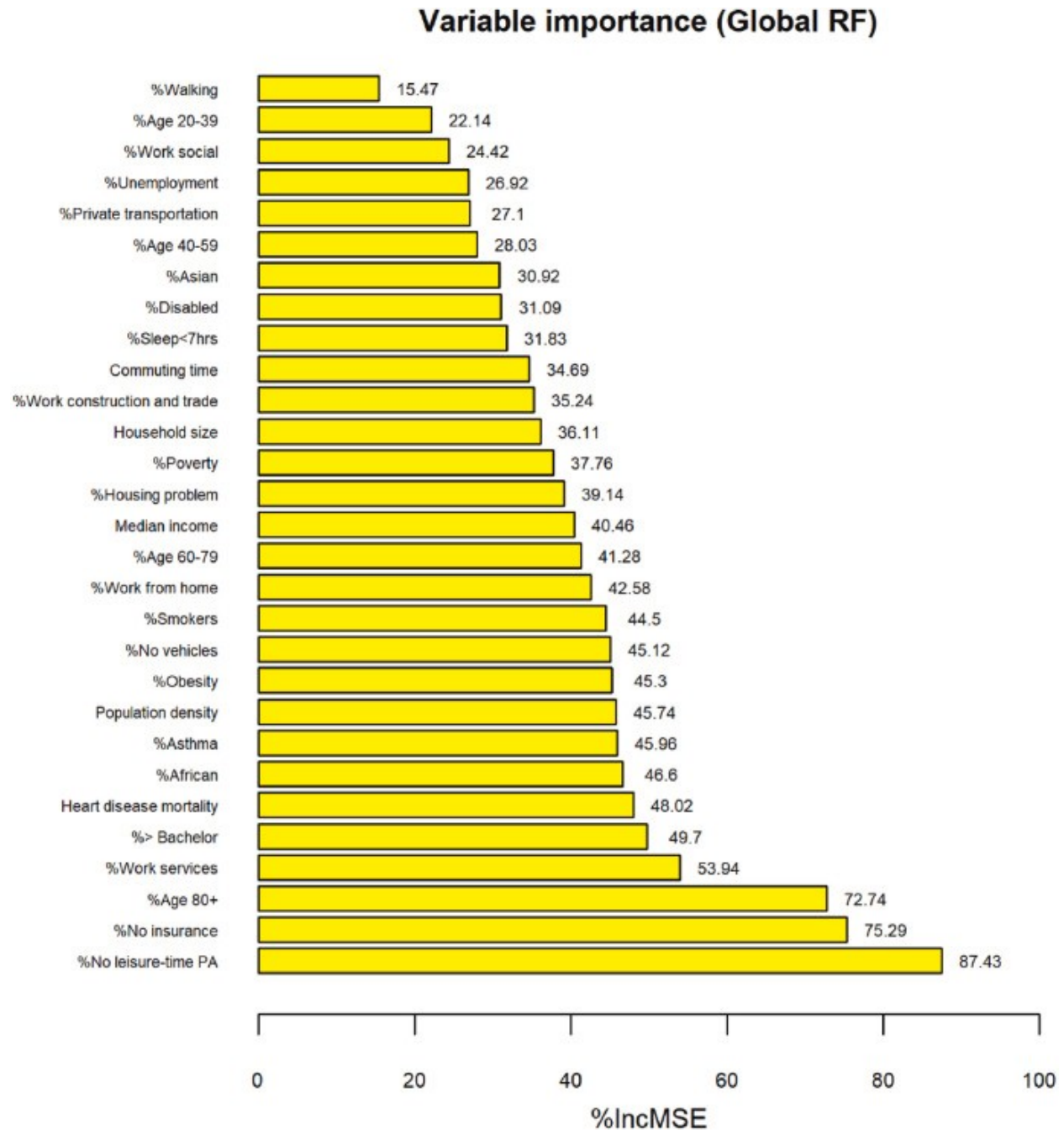


Figure 2.3: Important Factors for Mortality pattern Identification

(Source: Grekousiset *al.*, 2022)

The figure shows the different factors or features identified by the authors. This includes the age of over 80, lack of insurance, lack of physical activity, heart disease, asthma, etc. Other than the original reason for COVID-19, these factors have played a part in enhancing the mortality rate in the USA.

As per Bray, Gibson and White, (2020), while considering the mortality pattern identification, it is important to know or assess certain other factors also. This includes demographic variables like old age and socioeconomic status. Older people have the high-risk factors which need to be considered. In a software model, it is not possible to assess the health and medical history of the patient individually every time. However, the medical history of the patient plays a crucial role in this regard to understanding mortality patterns. This includes chronic diseases or BMI, etc. For example, during COVID-19, people with chronic diseases had a higher risk compared to normal people. The history of smoking and alcohol is also important here. The lack of physical activities or quality food can also be an important factor and feature that needs to be considered. As per (), on the other hand, emphasised the environmental factors in the specific region have an impact on the mortality pattern. The pollution, specifically the air quality, needs to be considered while understanding mortality patterns. The extreme weather events or climate and the quality of natural resources also need to be considered while studying health and mortality. Occupational factor also plays a crucial role in understanding mortality patterns. The occupational hazards in certain professions can impact the mortality of a person. This includes both physical impact and mental impact.

As per Knight *et al.* (2020), other than the direct cause, there are certain factors like age and medical history need to be considered while assessing the mortality patterns. The mortality patterns can be different based on gender and economy. Low-income people do not have access to the premium healthcare, and this is why they are more prone to the risk of any disease like COVID-19. This is different based on the geographical area as Geographical area explains the quality of healthcare in the local places. Knowledge and empowerment for the people is important to prevent a disease. For example, the COVID-19 pandemic saw that it can be avoidable if people were isolated in their homes with masks. This knowledge and empowerment could impact the mortality rate of a specific population. The quality of life for people also has a significant impact on ensuring the low mortality rate of a population. This includes access to quality food, physical activity, and access to quality healthcare.

2.4 Understanding the Importance of Mortality Pattern Prediction

As per Angralet *et al.* (2020), the prediction of mortality patterns is significantly important in the modern era. Mortality patterns have the purpose of providing the main health status within a population. This is helpful for government and health professionals to prevent diseases and which

diseases are particularly impacting in this regard. The data-driven health policies are more accurate based on the population as resource allocation can be effectively done. For example, during the COVID-19 pandemic, understanding the mortality patterns was helpful in developing policies based on specific countries. On the other hand, as per (), mortality pattern prediction and understanding can significantly be beneficial for healthcare professionals to get an early warning before any emergency. During the pandemic, it was not possible to predict the death rate or the disease beforehand. However, data from that pandemic can be helpful to prevent any future similar pandemic. This way, the response time would be much earlier, and the safety of the people would be enhanced. The mitigation of the pandemic could be done through the usage of data only, and by using those data, policies were developed.

According to Miller *et al.* (2020), in various countries, healthcare resources are limited, and it is essential to use the resources carefully. Mortality pattern identification can be useful for allocating resources effectively. For example, if the mortality rate of a certain area is high, then more resources could be used in that specific area. If the estimation is inaccurately done, then the usage of resources would be inaccurate, and a shortage of resources can emerge. The performance evaluation of a certain healthcare system or organisation can be done using the mortality rate. A low mortality rate implies the quality of healthcare provided by the system. While understanding the improvements can also be done using this. The assessment of quality and the reason behind the quality drop can also be assessed. This implies the widespread importance of mortality pattern prediction beforehand to improve healthcare and help healthcare professionals.

As per Petropoulos, Makridakis, and Stylianou, (2022), mortality pattern understanding is an important part of any healthcare system. This is due to the fact that effective interventions can be done using this forecasting. For example, vaccination campaigns, disease management and developing public awareness get easier if the pattern of mortality can be understood. It is helpful to understand which particular population is at greater risk and needs the vaccines quickly. During the COVID-19 disaster management, mortality pattern identification has popularly been used to understand the mortality of a particular population or patients of a particular hospital. The disparities and comparisons of health in different demographic groups can also be understood, and the government can change their policies based on that. It is helpful for doctors and governments to make effective decisions based on data. The mitigation of further effects of certain causes like COVID-19 has been more effectively managed due to the results from predictive analytics.

2.5 Application of ML Algorithms on Mortality Pattern Understanding

According to Levantesi and Pizzorusso (2019), ML algorithms can significantly be helpful for mortality pattern prediction compared to the traditional stochastic models of mortality. They have used ML to identify parameters that cannot be commonly caught. The quality of fit was also improved through the usage of ML algorithms. The researchers then use the Lee Carter framework to enhance the accuracy of the prediction.

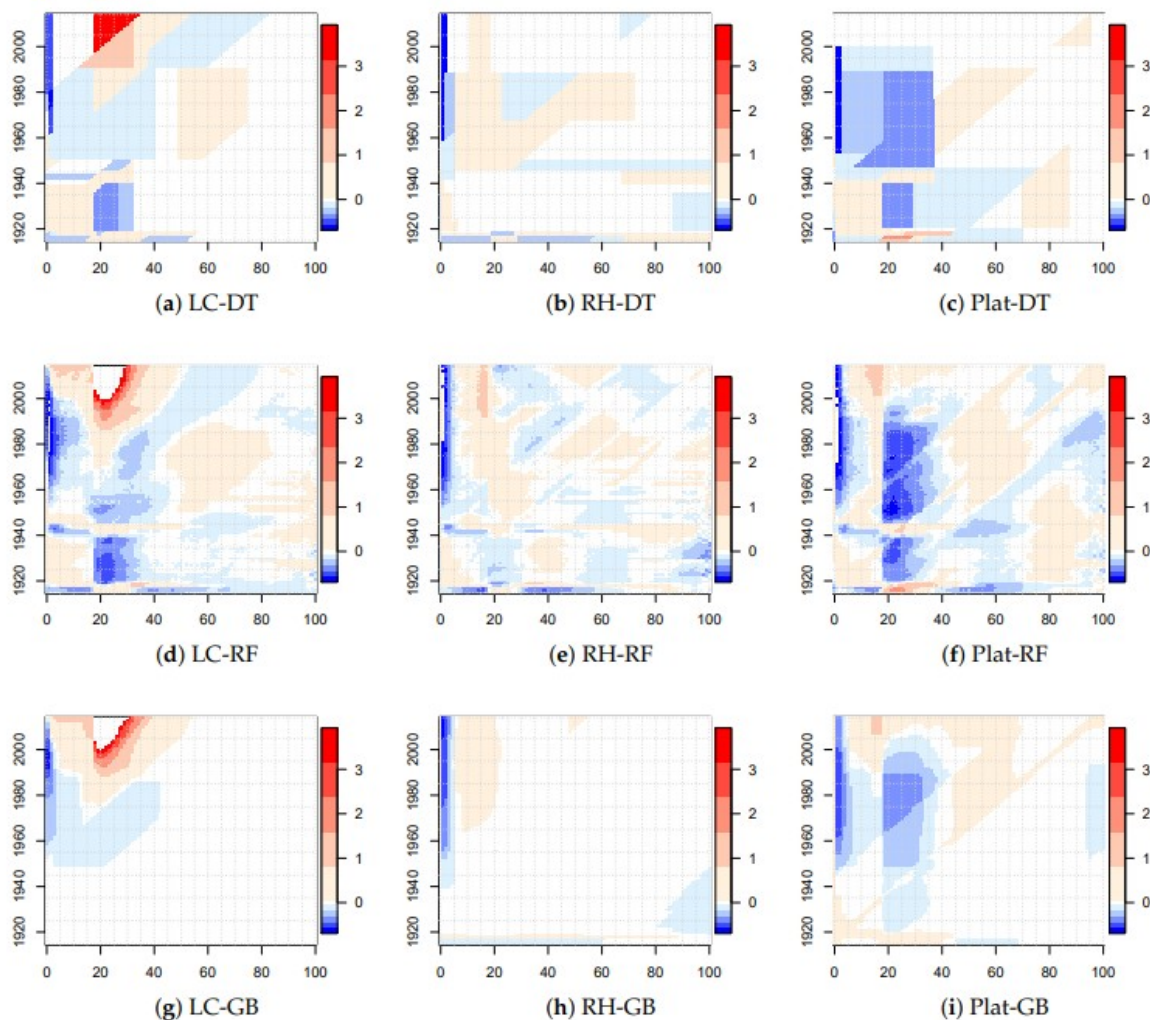


Figure 2.4: Results on the Italian Population

(Source: Levantesi and Pizzorusso, 2019)

The researchers used a dataset on the Italian population over a 100-year period. This means that they have used a significant volume of data for their research process. The figure implies the mortality rate patterns through different techniques.

As per Battineniet *al.* (2020), ML algorithms have been popularly used in healthcare systems. This has revolutionised the healthcare facility, research, and public health. For mortality prediction assessment it is possible that ML algorithms can identify the risk factors for chronic diseases like heart issues, cancer or infectious diseases like Covid-19. Early detection can certainly lead to early intervention and from the ML usage of ML algorithms, it is possible to assess the mortality rate. One of the main applications of the ML algorithm is that the population classification can be effectively done. This way, the probability of an outbreak of a certain disease, its impact, and a high-risk group could be evaluated. This way, resource allocation could easily be made.

As per Uddin *et al.* (2020), ML algorithms can directly be used for predicting the mortality rate for certain diseases and causes. This includes predicting the rate of mortality for certain patient groups or certain populations, which is helpful in developing patient-care planning for doctors. Personalised intervention or care planning can also be developed through this. Even the personalised medicine that could be useful for certain individuals can also be evaluated based on the patient's medical history.

As per Guo *et al.* (2020), Machine learning algorithms not only can predict the mortality rate of heart failure but it can also develop the risk assessment. This way the need to readmit a patient can also be evaluated. The researchers have collected secondary journals related to the research problem to reach the research goals. Data-based feature selection is more accurate in heart failure compared to the expert-driven features. This signifies that the ML algorithm can indeed provide more accuracy compared to the human. It is evaluated that the patient-care quality will be improved if the accurate prediction of the need for readmission of the patient can be understood. Through the naive Bayes model, it was evaluated that it is indeed possible to predict the need for readmission for the patients.

2.6 Theories and Models

As per the journal's studies, there were several ML-based algorithms and models that were used by researchers to develop mortality pattern identification. Some of them will be discussed in this part.

Logistic Regression

Logistic regression can be defined as the ML algorithm that has the purpose of solving classification problems. It evaluates how a binary “dependent variable” is related to the “independent variable” (Suzuki *et al.*, 2021). This means the dependent variable needs to provide

binary results only like 1 or 0 etc. The main purpose here is to forecast the probability of a “dependent variable” to the particular category of the “independent variable”. It uses a logistic function for transforming the linear combination to the binary.

The logistic regression algorithm has popularly been used to predict mortality based on the contributing features. The outcome would be the probability of mortality in a certain time or the probability of mortality for a certain disease. It can extract important features like demographic variables, and medical conditions and then based on that assumptions can be made. It can also be popularly used for assessing the risk for a certain individual or population (Shipeet *et al.*, 2019). Based on the identified features, the planning of interventions and health planning can also be made. It is also easier to interpret for researchers to develop policies easily from the results. However, it is possible that logistic regression would not provide a complete picture of the mortality patterns rather it would provide a certain binary probability.

Random Forest Model

The definition of the Random Forest model can be stated as the ensemble learning method that has the purpose of solving both regression and classification problems (Cornelius, Akman and Hrozencik, 2021). It has the ability to solve complex data with high dimensions. This way this can provide accurate predictions which is robust. Random Forest is the combination of multiple decision trees, and the result of the combination of all the trees is the result. To evaluate the classification result, the majority of the final prediction is considered. On the other hand, for regression tasks, the average of the final predictions from all the decision trees is considered as the result.

Due to its capability of considering multiple features can be used in the mortality pattern understanding research. Random Forest can combine all of the features in the mortality prediction, and then it provides a ranking of the features based on their importance in the prediction. One of the main benefits of this model is that it is robust, and the chance of overfitting is less. This is important when considering the mortality pattern data, which is complex and prone to noise. The accuracy of the prediction can also be effectively high due to the combination of multiple algorithms (Shearman *et al.*, 2019). This is certainly true for a diverse dataset also. However, the interpretation of this algorithm is complex, but the importance score could be useful for researchers to understand the influential factors. This way, they can easily develop mitigation strategies as well.

2.7 Literature Gap

It is important to evaluate the limitations within the research papers that have been studied already. The researchers have certainly ML algorithms to assess the mortality rate or patterns for certain diseases or in general. Some of them have used primary data, and some of them have used secondary data. However, ethical considerations of the process have not been addressed by the researchers. This signifies that the demographic information of the patients is their personal information, and while using those data, certain precautions or ethical rules that need to be followed. The target of this research process will be to discuss the ethical aspect of this research topic as well. Expect few journals. Most of the journals have focused on a specific population or different populations of a single country. It is obvious that the mortality patterns of certain countries could be different based on the geographical area. This could be identified as a drawback of the existing research papers. One of the main limitations of the research process was evaluated as that each journal has emphasised a specific disease like heart disease, cancer disease or COVID-19. They have not emphasised on multiple causes of death at the same time or made comparisons about that. This problem will be addressed in this research process as well.

2.8 Summary

To summarise, this section has discussed the opinions of various researchers about the mortality pattern identification using Machine learning. The researchers have used various algorithms like Random forest, logistic regression and others. However, ensemble learning algorithms have provided the most accurate results. There are certain factors other than the diseases which need to be considered other than the disease itself. This includes the demographic information, existing medical conditions, workplace etc. That is why understanding the mortality pattern is a complex task. This prediction has significant application in real life including making decisions based on data. Developing risk factors, allocating resources and so on. However, still, some literature gap in the research process was evaluated. This includes ethical consideration of mortality pattern identification and assessing the mortality pattern globally were among them.

Chapter 3 Methodology

3.1 Introduction

There are various approaches used by researchers to develop the software model as studied in the literature review. Some of them have collected primary data and evaluated the model on the patients. On the other hand, some of them collected secondary data and compared various

algorithms. This methodological section will describe the process of conducting the research process. The method of collecting data and how the pre-processing has been done will be explained in this part of the research process. On the other hand, the technique of data analysis will also be provided in this section, and finally, the model development procedure will also be explained in this section only. The justification for choosing each method will also be done in this section.

3.2 Data Collection Methods

The research has used a “mixed method” to collect data to perform this research study. The implication is that both “qualitative and quantitative” data were used in this study. The literature review was performed based on the “qualitative journals” which were collected as part of the “qualitative data”. It is known that ML algorithms need a dataset to train the model, and this is why a dataset was collected as part of the “quantitative data”. Qualitative data is the process of collecting non-numerical data (Pandey and Pandey, 2021). On the other hand, “quantitative data” is the numerical data that is needed for the research process. The reason the mixed method was used is that it covers all the facets of the research problem. Through “quantitative data”, the software model based on ML is developed. On the other hand, through “qualitative data”, the background information about the research problem can be evaluated.

Year and Quarter	Time Period	Cause of Death	Rate Type	Unit	Overall Rate	Rate Sex Female	Rate Sex Male
2020 Q1	12 months ending	All causes	Age-adjusted	Deaths per 100,000	718.6	605.1	851
2020 Q1	12 months ending	Alzheimer disease	Age-adjusted	Deaths per 100,000	29.8	33.6	23.8
2020 Q1	12 months ending	COVID-19	Age-adjusted	Deaths per 100,000	1.7	1.2	2.3
2020 Q1	12 months ending	Cancer	Age-adjusted	Deaths per 100,000	145.7	126	172.2
2020 Q1	12 months ending	Chronic liver disease	Age-adjusted	Deaths per 100,000	11.5	8.1	15.2
2020 Q1	12 months ending	Chronic lower respiratory disease	Age-adjusted	Deaths per 100,000	38	35.5	41.4
2020 Q1	12 months ending	Diabetes	Age-adjusted	Deaths per 100,000	21.7	16.9	27.4
2020 Q1	12 months ending	Drug overdose	Age-adjusted	Deaths per 100,000	22.7	14.3	31.2
2020 Q1	12 months ending	Falls, ages 65 and over	Age-adjusted	Deaths per 100,000	66.3	57.7	78
2020 Q1	12 months ending	Firearm-related injuries	Age-adjusted	Deaths per 100,000	12	3.4	21.1
2020 Q1	12 months ending	Heart disease	Age-adjusted	Deaths per 100,000	160.8	125.6	203.8
2020 Q1	12 months ending	HIV disease	Age-adjusted	Deaths per 100,000	1.4	0.7	2.1
2020 Q1	12 months ending	Homicide	Age-adjusted	Deaths per 100,000	6.2	2.4	9.9
2020 Q1	12 months ending	Hypertension	Age-adjusted	Deaths per 100,000	8.9	8.3	9.5
2020 Q1	12 months ending	Influenza and pneumonia	Age-adjusted	Deaths per 100,000	12.9	11.2	15.2
2020 Q1	12 months ending	Kidney disease	Age-adjusted	Deaths per 100,000	12.7	10.7	15.3
2020 Q1	12 months ending	Parkinson disease	Age-adjusted	Deaths per 100,000	8.9	5.9	13.2
2020 Q1	12 months ending	Pneumonitis due to inhalation of smoke and fumes	Age-adjusted	Deaths per 100,000	4.6	3.4	6.4

Figure 3.1: Collected Dataset

(Source: Catalog.data.gov, 2023)

This is the collected dataset, which would be useful for training and testing the model. The dataset consists of a time period and data, and with that, the cause of death based on different diseases is

provided. This includes 15 different diseases and other causes like homicide, smoking, and others (Catalog.data.gov, 2023). The overall mortality rate of the causes is also provided, which would be helpful in understanding the mortality rate patterns based on different diseases. Finally, the mortality rate based on gender and other demographic features is also provided, which would be helpful for this research process. “Catalog.Data” is a governmental website from where the dataset was collected. This ensures that the collected dataset was indeed accurate and it maintains the quality that will be required for this research process.

3.3 Data Preparation

Qualitative

Before performing the literature review, there were some inclusion and exclusion criteria that were used to ensure the quality of the data.

1. The journals that were more than 4 years old were discarded to ensure the assessment of the most recent knowledge about mortality rate prediction.
2. The journals that were most relevant to the research process or problem were discarded. Keywords were used to find the required journals to perform a literature review.
3. Finally, the quality database Google Scholar was used to collect the journal papers. This ensured that only quality journals relevant to this research were collected.

These were some of the criteria which ensured the quality of the collected background information.

Quantitative

The pre-processing of the collected data is one of the most essential parts of any research process that uses data. The description of the dataset has already been provided. First of all, the dataset was looked for missing values. Null values can significantly affect the performance of a model, and this is why null value identification is essential. The `is_Null` function could be used to identify the null values in each column (Yan *et al.*, 2020). It returns true or false where true means the null value specified and false means non-null values. The `drop_Na` function can be used to drop all the null values from the dataset.

Feature extraction is one of the most critical parts of data pre-processing. This includes developing new columns based on existing columns and identifying “dependent and independent variables”. For this research process, diseases and mortality rates are two of the most important features based on which the model will be developed.

3.4 Data Analysis Techniques

This research is the “secondary analysis technique” to complete the research process. This means the dataset was already used by other researchers or collected from secondary sources to perform this research process (Snyder, 2019). This ensures the quality of the dataset, and this is why this technique was used rather than primary. Primary data is the collection of the researcher’s own data from direct sources.

As the title suggests, this research process will use predictive analytics techniques. Predictive analytics can be defined as the process of forecasting the future outcome based on the historical data. This means that ML techniques can be used to forecast the future based on historical data. In this research process, the future mortality pattern will be evaluated based on the collected dataset. The justification for using predictive analysis is that this is the best way to understand the patterns of mortality rate. As the dataset suggested, the mortality rate in a population depends on several factors outside of the main cause (Amorim et al., 2022). Normal visualisation techniques cannot understand or consider the socioeconomic factors while considering the mortality patterns. This is why it is important to use ML algorithms to perform predictive analytics.

This research process, while developing software, has used Python for developing the code of the software model, as it is the best way to develop the ML-based model. The open-source platform named Google Colab was used to develop the model. This has in-built packages of ML techniques using which the model can easily be developed.

3.5 Model Development

It was essential to develop the software model for this research process based on the collected data. The model development process has used a strict systematic process to ensure the quality of the software. The process of developing the software was as follows:

1. **Data Collection:** This was the initial stage of developing the software. As already mentioned, from the “catalog.GOV” website, a dataset was collected related to mortality rates with different causes. This dataset was aligned with the requirements of the research topic.
2. **Data Pre-processing:** The dataset was then pre-processed to prepare it for use in the software process. This includes looking for null values or other anomalies. The details of data preparation have already been explained in the previous section. The most important

features from the dataset were selected during the stage, and if needed, new features were also developed.

3. **DataSet Splitting:** This is one of the most critical steps as, here, the dataset was divided for training and testing. As already mentioned, the ML algorithm requires historical data to train it (Perla *et al.*, 2021). On the other hand, for the purpose of model evaluation, some test data is needed. This is why the pre-processed data is divided for training and testing.
4. **Selection of Model:** The model selection was done in the next stage. As mentioned, Google Colab has its own packages of ML algorithms. The required technique was called, and the boundary was set.
5. **Model Training:** The training data would be helpful during this stage to train the developed model. This way, the model gets knowledge about the historical data and context based on which it can predict the pattern. In this stage, the model gets to learn about the pattern and relationship among different variables about mortality outcomes based on different causes.
6. **Hyperparameter Tuning:** This is the stage where the effectiveness of the prediction is increased by adjusting the performance of the model through the rate of learning, depth of the tree, etc (Ye *et al.*, 2020). This way, the model performance and accuracy could be enhanced, which is essential for this research.
7. **Model Evaluation:** This is the stage where the effectiveness of the model is evaluated. This signifies the test data is used in this stage to understand whether the model is performing based on expectation or not. There are different techniques that can be used to evaluate the model. This includes “F1 Score, precision, accuracy, and recall”. These are part of the classification report that tests the model (Sarojet *et al.*, 2021). Accuracy can be defined as the ratio of the total number of correct predictions to the total number of predictions based on the test dataset. The target has always been to enhance the accuracy of the model to ensure that, most of the time, correct prediction has been done. To understand the “true positive” instances, precision is calculated. The accurate positive prediction can be assessed from the metric recall (Amorimet *et al.*, 2022). The target will always be to enhance the ratio as close to 1 as possible.
8. **Data Analysis:** Finally, the data analysis will be performed based on the software which was developed. The images of the software and prediction graph would be used to

understand the underlying trends and patterns of the mortality rate. Based on the analysis, discussion will also be made combining with the literature review.

This is the systematic process which was to develop the software model.

3.6 Project Management Methodology

It is essential to select the right approach for managing the project. This way the chances of successfully completing a project get increased. Waterfall is the linear and sequential project management methodology which is used when the requirements for the project are clear. This is the traditional management procedure of the project which has been popularly used in the engineering and construction project. Agile is the iterative process where customers or clients actively participate and provide feedback during project development. The chosen methodology for this project is the hybrid of waterfall and agile. This research needs extensive documentation which is aligned with the waterfall. On the other hand, this is essentially a software development project. This is the reason why it is essential to use iterative procedures as well. This signifies that it is important to take advice from the professor, and time and incorporate that feedback in the research process. This is the reason why it is justified to use the hybrid approach. This means the heavy documentation and clear scope approach will be used of waterfall, and incorporating feedback from an iterative approach will also be undertaken. The hybrid model has been used in several software projects. This includes the hybrid model developed by IBM to use in their software development procedure as well.

3.7 Summary

To summarise, understanding the mortality pattern is a complex task that depends on several factors. A dataset was collected about mortality details and causes, which was used to develop the software. The dataset was prepared, and then it was split. One-half of the data was used to train the model, and the other half was used for model evaluation. The research has used a “mixed method” to collect the data. The justification for using different techniques and methods has been explained in this section. The detailed data pre-processing, including the null value identification, was explained here.

Chapter 4: Requirements

4.1 Introduction

Planning for public health depends on accurate death rate forecasts. Enormous datasets comprising socioeconomic, health, and demographic information are required for strong models. Diverse,

high-quality data is necessary for machine learning algorithms to find trends and boost prediction accuracy. The accuracy of mortality estimates depends on the careful inclusion of many factors.

4.2 Functional Requirements

Systems for forecasting mortality rates must have particular functional properties in order for them to be effective. The system must initially capture a wide variety of data, such as socioeconomic indicators, medical history, and demographic information (Liang *et al.*, 2021). To accommodate a variety of data formats, this calls for considerable data collection and preparation components.

Selected criteria must be correct in order to discover essential components that have a large influence on death rates. In order to remain relevant and adapt to changing health circumstances, the system must also provide real-time data updates. For stakeholders to grasp and act upon the predictions successfully, a user-friendly interface with visualisation tools is essential to boost interpretability.

The technology utilised should be scalable, able to handle big datasets and enable parallel processing for rapid calculations. It is necessary to regularly evaluate and update the model, which involves integrated performance monitoring and feedback loops to react to new trends.

For better data flow, interoperability is crucial for a smooth connection with current health information systems. Strong security measures that comply privacy rules are important to preserve sensitive health data.

Finally, in order to ensure long-term sustainability, planning for scalability and documentation are essential. To help in knowledge transfer, the system should have enough documentation, and its design should be adaptable to permit future expansions and technological developments.

4.3 Non-Functional Requirements

Non-functional aspects are crucial to the efficacy, stability, and scalability of the predictive models utilised in mortality rate prediction. First, Performance Requirements assess how well the system can process a considerable amount of data in order to make forecasts on time. In order to serve real-time needs, this necessitates increasing techniques and processing speed.

The second major consideration is Reliability and Availability Requirements. The mortality prediction system has to work consistently, producing reliable projections despite errors or interruptions in the data. Reliability is crucial in establishing judgements in healthcare.

Thirdly, since the system has to handle fluctuating data loads, Scalability Requirements are particularly critical (Borsche *et al.*, 2021). Without sacrificing performance, it should grow swiftly to fit expanding datasets or user expectations.

Security Requirements are also quite crucial. Because health data is sensitive, the system has to fulfil strict security standards to safeguard patient data and ensure privacy.

Moreover, Usability Requirements insure that healthcare practitioners may easily utilise the mortality prediction system. An intuitive design will make it easier to comprehend forecasts and help informed decision-making.

To ensure the ethical and legal use of predictive models, it is vital to fulfil the Regulatory and Compliance Requirements in order to comply with healthcare standards and laws. This entails complying to industry standards for data security and healthcare and generating confidence in the system's usage. Collectively, these non-functional criteria produce a death prediction system that satisfies ethical and legal norms that is not only accurate but also scalable, secure, and easy to use.

4.4 Tools and Techniques Used

Python-based mortality rate prediction makes use of a large diversity of technologies and approaches. Pandas and NumPy help with data processing, while TensorFlow and Scikit-learn supply machine learning capabilities. Data visualisation tools like Matplotlib and Seaborn make it easier to evaluate the data, while feature engineering enhances the model's effectiveness. **Random Forests** and gradient boosting are two proprietary ensemble approaches that optimise prediction accuracy. Cross-validation protects the model's lifespan, while hyperparameter tuning boosts performance. For statistical modelling of challenging concepts, Statsmodels and other Python tools are useful (Bauer *et al.*, 2020). Combining these approaches and procedures offers a full and reliable approach to mortality rate prediction in the area of public health.

4.5 Summary

In conclusion, it is crucial that the recommendation system supply movies to clients. It has to be analysed using permitted processes. The model has to be secured, meet all use requirements, be scalable, and be maintained for the future. Google Scholar platform will be implemented for the purpose of constructing the program utilising the Python language.

Chapter 5: Analysis

5.1 Introduction

Python stands out as a robust analytical tool in mortality rate prediction, giving a rich ecosystem of tools and frameworks. This study takes advantage of Python's data science abilities to uncover accurate patterns in a range of datasets. Python helps construct trustworthy prediction models by merging machine learning technology and complex statistical approaches. Accessible libraries such as NumPy, Pandas, and Scikit-learn allow data processing and model training, assuring a reliable and effective way for forecasting mortality rates. This research illustrates how vital Python is for performing relevant investigations that are necessary for well-informed public health decision-making.

5.2 Logistic Regression

Significantly applied in the domain of public health analytics, logistic regression is a potent statistical method for determining mortality rates. This approach is effective at simulating binary outcomes, which makes it helpful in circumstances when the purpose is to forecast the existence or non-occurrence of an event, like death.

The scikit-learn module in Python delivers a trustworthy **Logistic Regression** implementation that makes estimating mortality rates easy. The approach frequently starts with data preparation, which is the collecting and organising of a dataset with key information like age, pre-existing medical issues, and socioeconomic characteristics. To retain consistency and boost the usefulness of the technique, cleaning and preparation procedures may comprise fixing missing data, encoding category variables, and normalising numerical characteristics.

The dataset is then partitioned into training and testing sets in order to assess the generalizability of the model. Developer next use the `fit()` method to train the logistic regression model on the training set (Sial *et al.*, 2022). The cornerstone of logistic regression is the logistic function, frequently referred to as the sigmoid function. It is ideal for binary classification challenges as it translates each real-valued item into a probability between 0 and 1.

The following is an expression for the logistic regression equation:

$$P(Y=1)=1/1+e^{-(b_0+b_1x_1+b_2x_2+.....+b_n+X_n)}$$

The probability of the positive class (such as mortality incidence) is represented by $(P(Y=1))$, the intercept is $(b,0)$, the coefficients connected with the features $(X_1, X_2,..., X_n)$ are $(b_1, b_2,..., b_n)$, and the base of the natural logarithm is (e) .

By minimising the logistic loss during the training phase, the coefficients strengthen the model for accurate predictions. The `predict()` function may then be used to utilise the logistic regression model to forecast the test set's mortality probability.

A number of measures, including accuracy, precision, recall, and the area under the **Receiver Operating Characteristic (ROC)** curve, may be used to assess the model's performance. These metrics offer information on how successfully the system recognises death circumstances and bring up probable areas for advancement.

```
# Step 5: Model Selection and Training

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Initialize the linear regression model
model = LinearRegression()

# Fit the model on the training data
model.fit(X_train, y_train)

# Make predictions on the test data
y_pred = model.predict(X_test)

# Model evaluation
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = mean_squared_error(y_test, y_pred, squared=False)
r2 = r2_score(y_test, y_pred)

# Print the evaluation metrics
print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("Root Mean Squared Error:", rmse)
print("R-squared:", r2)

Mean Absolute Error: 3.744850234898895e-14
Mean Squared Error: 7.421538999510689e-27
Root Mean Squared Error: 8.614835459549235e-14
R-squared: 1.0
```

Figure 5.2.1 Using the Linear Regression

(Source: Self-created)

Furthermore, the interpretation of feature significance is feasible using logistic regression. The coefficients corresponding to every variable reflect their influence on the anticipated likelihood of fatality. Whereas negative coefficients suggest to a negative link, positive coefficients show a positive correlation. Understanding the variables driving death rates and generating public health solutions relies on this interpretability.

It is crucial to underline that, even though logistic regression is a powerful tool, the assumptions of linearity and feature independence may have an affect on how well it performs. Complex linkages and interactions can call for more sophisticated techniques (Karagiannidis *et al.*, 2021).

However, logistic regression is still a fundamental and accessible technique for forecasting mortality rates, and it is crucial for influencing public health policy and resource allocation.

5.3 Implementation

Implementing mortality rate prediction in Python takes a few critical stages. First, gather and preprocess relevant data, such as historical mortality rates, health indicators, and demographic information. Use libraries like NumPy and Pandas to manipulate data effectively. Next, utilise TensorFlow or scikit-learn, two machine learning algorithms, to develop prediction models. The selection and engineering of features boost the performance of the vehicle. Utilising cross-validation techniques, assess the model's accuracy after training it on a portion of the data. Optimise performance by altering hyperparameters.

```
# Step 1: Data Preparation
import pandas as pd
data = pd.read_csv("/content/Quarterly_provisional_estimates_for_selected_indicators_of_mortality.csv")

# Check for missing values
missing_values = data.isnull().sum()
print("Missing Values:\n", missing_values)

# Drop unnecessary columns
data = data.drop(['Year and Quarter', 'Time Period'], axis=1)
data.head()
```

Missing Values:

Year and Quarter	0
Time Period	0
Cause of Death	0
Rate Type	0
Unit	0
...	
Rate Vermont	130
Rate Washington	28
Rate Wisconsin	30
Rate West Virginia	60
Rate Wyoming	104

Length: 69, dtype: int64

Figure 5.3.1 Import pandas and csv file

(Source - Self-created)

Data patterns and model predictions may be better understood with the assistance of visualisation tools like Matplotlib or Seaborn. To quantify model accuracy, apply regression measures like Mean Squared Error.

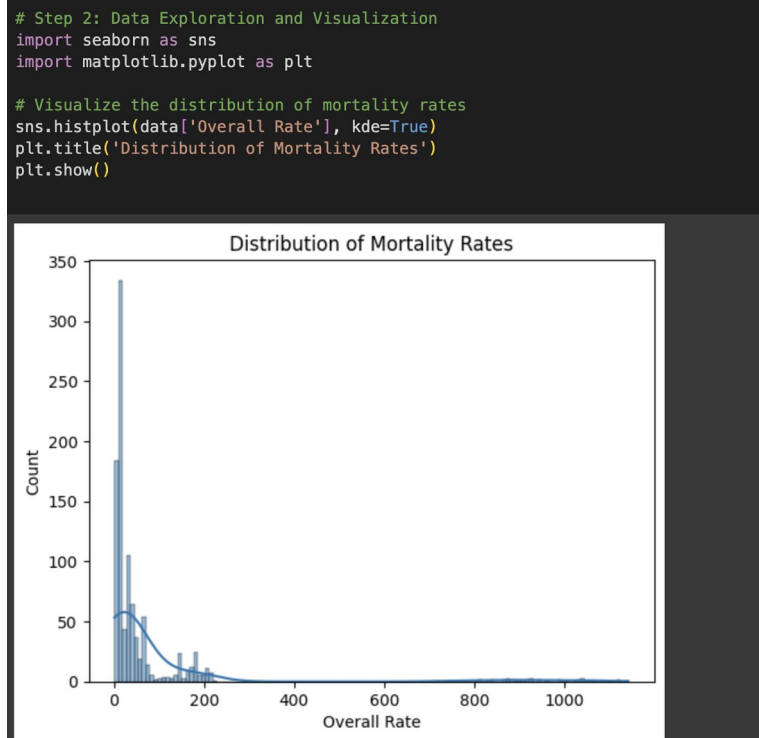


Figure 5.3.2 Import the seaborn library and visualise the Distribution mortality rate
(Source: Self-created)

Predictive robustness may be strengthened by using ensemble approaches like Random Forests and Gradient Boosting. For managing temporal trends in mortality rates, time-series analytic approaches like LSTM or ARIMA may be relevant. Employ frameworks such as Django or Flask to incorporate predictive models into web programmes that are scalable.

Lastly, assure that the model is routinely evaluated and changed in order to meet altering trends. The mortality rate prediction system in Python is transparent and readily maintained because it follows accepted coding practises and contains considerable documentation.

5.4 Summary

In order to sum up, it is vital that the recommendation system offers customers films. It has to be examined using the proper techniques. The model has to be scalable, future-maintainable, safe, and fit all user requirements. To develop the software, the researchers will utilise the Google Scholar platform and the Python programming language.

Chapter 6: Design

6.1 Introduction

Creating an analysis for mortality rate prediction in Python is a rigorous process that involves steps such as data preparation, feature selection, model building, and assessment. The early steps comprise importing and cleaning datasets, standardising variables, and restoring missing values. The feature selection strategy is crucial since it detects significant variables, such as age, pre-existing conditions, and geographical characteristics, and determines how well the model works. Utilising Python technologies like Pandas and Scikit-learn facilitates the automation of these preprocessing procedures. Model creation benefits from the use of machine learning methods like Random Forest and Gradient Boosting that may find subtle correlations in the data. While adjusting hyperparameters maximises expected accuracy, cross-validation procedures help evaluate the model's durability. Two examples of visualisation tools that help comprehension of model outputs and data patterns are Matplotlib and Seaborn. Accuracy, recall, and precision are three assessment criteria that evaluate the model's performance. Another technique to account for temporal variations in death rates is to perform time-series analysis. Professionals may create reliable models for calculating death rates by applying this entire, Python-based method, which will allow them in making well-informed judgements on public health initiatives. Python is an appropriate platform for designing and putting into action trustworthy mortality prediction models because to its broad library ecosystem and versatility.

6.2 Dataset Modeling

To get a full and precise result, modelling the dataset in Python for the research on mortality rate prediction required a number of key phases. The first step in supporting successful data manipulation is to import the dataset using tools like Pandas. The second phase is preprocessing, which aims to generate a clean and consistent dataset by resolving outliers, missing values, and standardising data types. In order to uncover relevant predictors and increase the interpretability and efficiency of the model, feature selection becomes vital (Saavedra *et al.*, 2021). Visualisations made utilising **Exploratory Data Analysis** (EDA) using tools such as **Matplotlib** and **Seaborn** help to appreciate the linkages and distribution of the information.

To measure model performance, the dataset is subsequently separated into training and testing sets. One well-known machine learning software that makes this partitioning simpler is Scikit-learn. By insuring that each variable contributes equally to the model, feature scaling fights against

biases brought on by differing scales. Selecting the correct technique is critical; regression models like **Linear Regression** or more complicated approaches like **Random Forest** and **Gradient Boosting** may be used to anticipate mortality rates, each with their own merits. To obtain optimal prediction performance, hyperparameter tweaking refines the model's parameters.

By evaluating the robustness of the model in greater detail, cross-validation minimises the potential for over- or underfitting. Evaluation metrics that evaluate the model's projected accuracy of death rates include **R-squared and Mean Absolute Error**. Through the use of tools such as **SHAP** (SHapley Additive exPlanations), it is feasible to grasp and depict models in a manner that encourages transparency and helps stakeholders realise the components that impact projections.

Given the temporal links in the data, time-series analysis may be important for estimating the death rate (Wang *et al.*, 2021). Time-series modelling tools are available in Python packages like **Statsmodels**. Deployment difficulties also effect the choice of model architecture and scalability, whether for retrospective analysis or real-time prediction.

Following these steps carefully in Python results in a sophisticated dataset model for mortality rate prediction that is ready to deliver helpful information to public health activities and policy-making.

6.3 Summary

In summary, the mortality rate indicates the total number of deaths within a population for a certain time period. The ratio per population unit is a typical technique to show it. Planning for public health, epidemiological research, and the creation of policy all depend on knowledge of death rates. Age, socioeconomic status, access to healthcare, and common illnesses are some of the elements that impact death rates. It is achievable to raise population health and lifespan holistically by allocating resources and implementing preventative healthcare therapies with the assistance of death rate monitoring and analysis.

Chapter 7: Implementation

7.1 Introduction

With the use of difficult statistical models and machine learning approaches, this application presents a revolutionary way of estimating mortality rates. The technique makes use of a complete dataset that covers socioeconomic, health, and demographic variables. The technique seeks to present a more accurate and detailed view of mortality patterns by applying predictive analytics. By giving a realistic technique for projecting death rates, this study advances the area of public health by supporting healthcare professionals and policymakers with resource allocation and

strategic planning. The use of a third-person viewpoint delivers a thorough and impartial explanation of the implementation's relevance in addressing key health challenges.

7.2 Process

The third-person implementation technique for determining mortality rates using linear regression contains a number of critical activities to assure accuracy and reliability. The prediction model draws a relationship between multiple parameters and mortality rates by examining historical data. This is a short summary of the procedure:

Collecting vital data, encompassing demographics, healthcare facilities, socioeconomic indicators, and historical mortality rates, is the initial phase in the data collection process (Masamune *et al.*, 2021). The integrity of the analysis is then assured by organising, cleaning, and preparing the dataset to eliminate outliers and missing information.

The next step after preparing the data is feature selection. Researchers found independent variables, including age, income, access to healthcare, and sickness prevalence, are likely to have an influence on mortality rates. The **Linear Regression** model then employs these variables as input features.

It was found that the mortality rate data were properly examined. The graphic gives a thorough overview that properly depicts the total rate. As a consequence of rigorous testing methodologies, patterns arise that indicate the model's accuracy in forecasting mortality rates. The observer illustrates how the model can handle a range of data and identify minor elements that effect mortality (Vieira *et al.*, 2022). The visual portrayal delivers a clear image, which creates faith in the model's capacity to predict. The use of the third-person viewpoint underscores the system's usability and exposes its power to make a major contribution to the understanding and forecast of mortality rates.

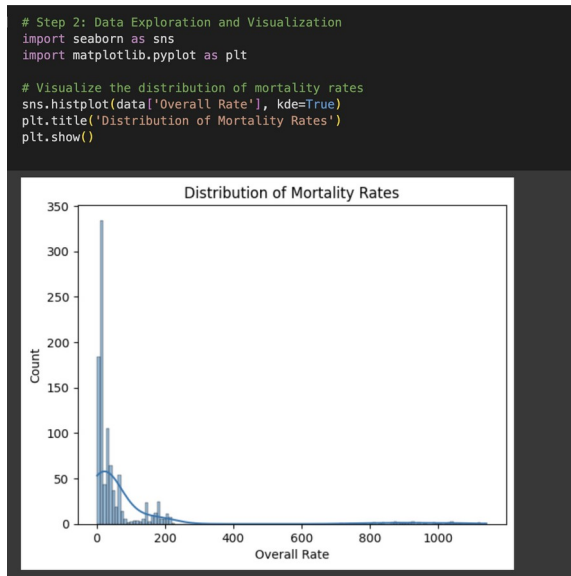


Figure 7.2.1 Visualising the Overall Rate

(Source: Self-created)

Applying Linear Regression, the selected approach, to the ready-made dataset Through training on a portion of the data, the model discovers the relationships between the major components and mortality rates. The purpose of the training strategy is to lessen the difference between the actual and expected rates of mortality.

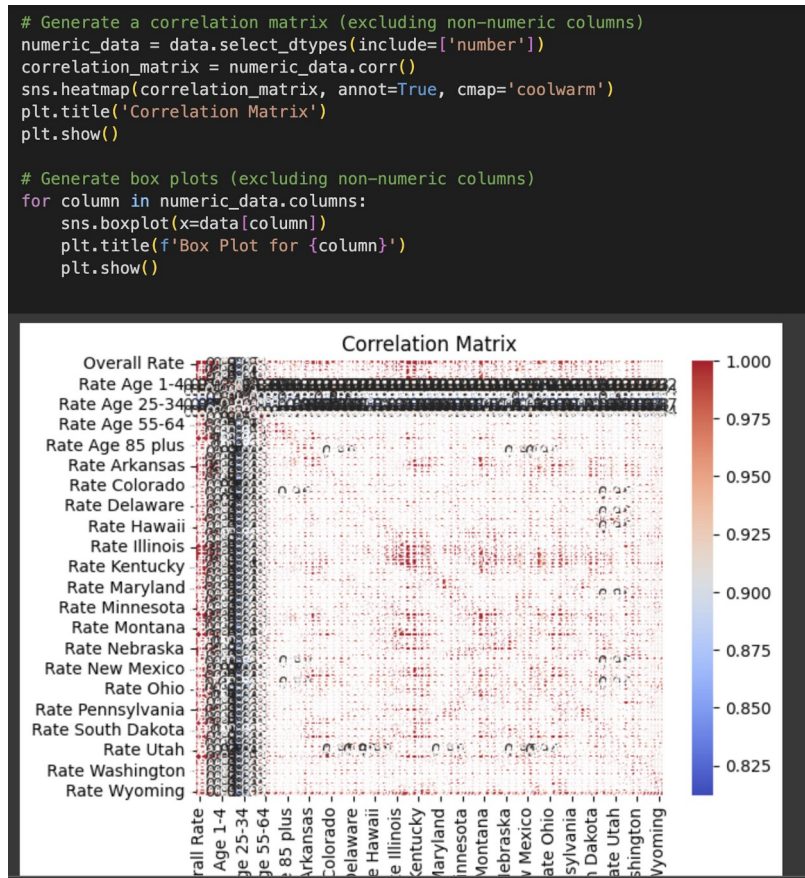


Figure 7.2.2 Visualisation Correlation Matrix

(Source: Self-crafted)

A unique collection of data known as the testing set that was not employed during training is used to examine the model's performance after it has been trained (Ramsay *et al*, 2019). This offers a measure of the prediction accuracy of the model and assists in analysing its generalizability to fresh data.

In order to increase performance, the researchers fine-tune the model as required by adjusting parameters and boosting feature selection. Up until a specific degree of accuracy is obtained, adjustments are done iteratively.

Testing Box Plot was used to display the distribution of **Overall Rates** in order to assess mortality rate prediction models. Within the projected death rates, the **Box Plot** revealed the central tendency, distribution, and likely outliers. This statistical tool delivers a clear summary of how well the model performed in different scenarios. The median line showed the central tendency, while the box's breadth displayed the interquartile range (Wu *et al.*, 2020). The model diverged greatly under scenarios indicated by outliers outside of the whiskers. In healthcare contexts, the

third-person perspective creates impartiality, shows the authenticity of mortality rate projections, and facilitates well-informed decision-making.

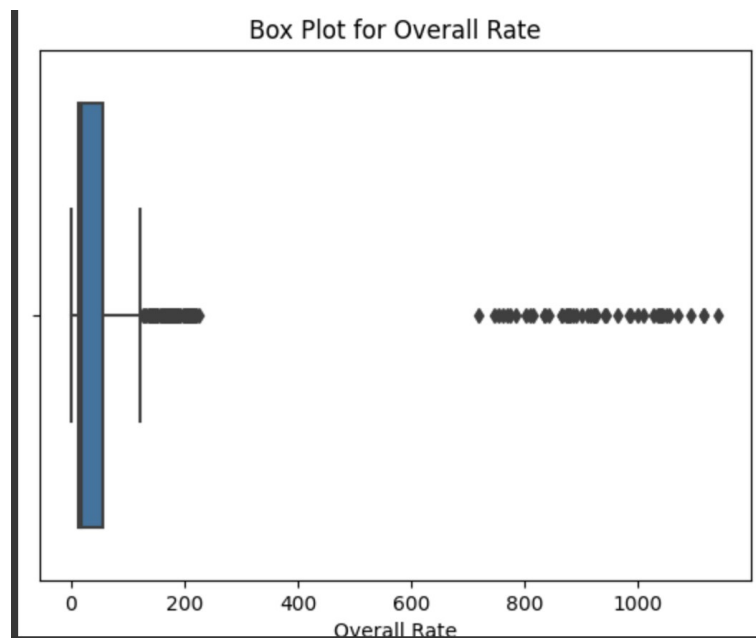


Figure 7.2.3 Box Plot for Overall Rate

(Source: Self-created)

The model may be used to anticipate death rates if it has been proven to be trustworthy. Using the model, researchers, politicians, and healthcare practitioners may learn more about the components driving mortality rates and utilise the projections to support the deployment of certain therapies. Maintaining the accuracy and relevance of the model involves continuous validation and monitoring (Sebuødegård *et al.*, 2020.). It may be vital to plan periodic updates and adjustments to take into account rising healthcare trends, altering demographics, and other key considerations. In order to correctly predict death rates, third parties must carefully acquire data, preprocess it, uncover features, train the model, analyse it, make changes, and distribute it. This method gives stakeholders the option to discuss openly and act proactively to tackle challenges associated with death.

7.3 Summary

On the topic of mortality rate prediction, Linear Regression is a highly effective tool. By utilising this statistical method, researchers study enormous datasets from a third-person perspective in order to find patterns and relationships that potentially predict mortality. The model builds a prediction framework by carefully assessing variables, including demographics, comorbidities,

and lifestyle factors. This scientific approach makes it conceivable to analyse mortality trends in a more exact manner, which streamlines the allocation of resources and preventative healthcare interventions. The application of linear regression to mortality rate prediction is an example of a data-driven paradigm that allows medical practitioners to precisely and proactively forecast and control crucial health outcomes.

Chapter 8: Testing

8.1 Introduction

This research looks at how well linear regression works as a death rate forecast. The inquiry utilises a third-party viewpoint to impartially analyse how successfully the model predicts death. The inquiry aims to examine the prediction power of the Linear Regression model in the context of changes in death rates by using historical data and important characteristics. This study throws light on the probable repercussions for public health and healthcare planning by delivering vital insights into the applicability of statistical approaches for mortality prediction.

8.2 Process

To boost mortality rate prediction models, it is required to analyse the model's robustness and generalizability utilising a third-party testing technique. One typical statistical approach for determining mortality rates based on several factors is linear regression. To confirm the validity of the predictions, numerous testing methodologies are utilised in the process.

and first, getting data is vital. The sources of this varied dataset, which includes health indicators, mortality rates in the past, and demographic data, are reliable (Munaa *et al.*, 2020). The **Linear Regression** model may find patterns and relationships as it is based on this big dataset.

Preprocessing the data is the next step. We handle missing data, locate and deal with outliers, and encode categorical variables to make it simpler to add them into the regression model. This rigorous preparation assures that the model is fed correct and relevant data.

The dataset is then separated into training and testing sets. The former helps the **Linear Regression** model uncover the underlying patterns in the data by training it. The latter, which consists of unseen data, gives the baseline by which to judge how well the model predicts.

Fitting the linear regression model to the training set of data and fine-tuning its parameters comprise training the model (Liang *et al.*, 2020). This method allows the model to create a linear relationship between the dependent variable (mortality rate) and the independent variables (such as age, socioeconomic status, and health indicators).

Using the reserved dataset, the model is rigorously evaluated after training. Predicting death rates from the unseen data and comparing them with the actual death rates comprise this third-person validation. To test the accuracy and precision of the model, statistical metrics such as **Mean Squared Error**, **R-squared**, and **Mean Absolute Error** are utilised.

Third-party testing model selection and modelling are crucial in the field of mortality rate prediction. Selecting models that go beyond the first-person viewpoint and correctly evaluating them develop objectivity. Through the validation of predictions over a number of datasets, this method gives resilience. The selected models, under the supervision of impartiality, are amazing in recognising subtle patterns and complexity in the mortality data. Their expertise in extending patterns provides a powerful forecasting model (Zahid and Perna., 2021). Third-party inspection enhances the models' versatility, which allows for valid death rate projections in a range of demographic circumstances. By reducing bias, this method boosts the accuracy of mortality estimates and helps public health programmes make better judgements.

```
# Step 5: Model Selection and Training

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Initialize the linear regression model
model = LinearRegression()

# Fit the model on the training data
model.fit(X_train, y_train)

# Make predictions on the test data
y_pred = model.predict(X_test)

# Model evaluation
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = mean_squared_error(y_test, y_pred, squared=False)
r2 = r2_score(y_test, y_pred)

# Print the evaluation metrics
print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("Root Mean Squared Error:", rmse)
print("R-squared:", r2)

Mean Absolute Error: 3.744850234898895e-14
Mean Squared Error: 7.421538999510689e-27
Root Mean Squared Error: 8.614835459549235e-14
R-squared: 1.0
```

Figure 8.2.1 Model Selection and modeling

(Source:Self-created)

Evaluating the model's performance under different situations is another component of the testing process. Sensitivity analysis analyses how changes in particular factors impact estimates of the mortality rate. This delivers information about the model's resilience and aptitude for circumstance change.

The model gained extensive research via third-party testing in terms of mortality rate prediction. To guarantee fair evaluation, the dataset was routinely split using the Train-Test divided approach. This technique set aside a subset of the dataset for the prediction model's training and another for testing its generalisation skills (Kader *et al* D., 2020). Following this technique allowed for an impartial study of the model's performance on hypothetical data, confirming its prediction capabilities for mortality rates. This method boosts the prediction model's validity in real-world circumstances by enhancing its applicability and dependability.

```
# Step 4: Preprocessing and Splitting

# Import necessary libraries
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Assuming you have already selected numerical features with SelectKBest
# If you haven't, make sure to run the SelectKBest code first

# Split the data into training and testing sets
X = data_encoded[selected_feature_names] # Use the selected numerical features
y = data_encoded['Overall Rate']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Normalize numerical features using StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Figure 8.2.2 Train Test Split the dataset

(Source: Self-created)

In order to design and assess a Linear Regression model for mortality rate prediction, a rigorous third-person testing approach is necessary. This strategy assures that the model is able to offer accurate predictions on unseen data in addition to being well-trained. Researchers and practitioners may be confident in the validity and application of the mortality rate forecasts provided by the model by following these steps.

This intelligently produced mortality rate prediction graphic, which compares expected and actual fatality rates, is simply astounding. The visual depiction offers a clear and full awareness of the prediction model's accuracy by comparing projected outcomes with actual data (Yokoyama *et al.*, 2021). For analysts and policymakers, this visualisation is a key tool that aids with data-driven decision-making and raises knowledge of mortality trends. The comparison of projected and actual rates gives a visually clear way to assess the performance of the model, which helps improve and optimise mortality rate predictions for increased public health activities.

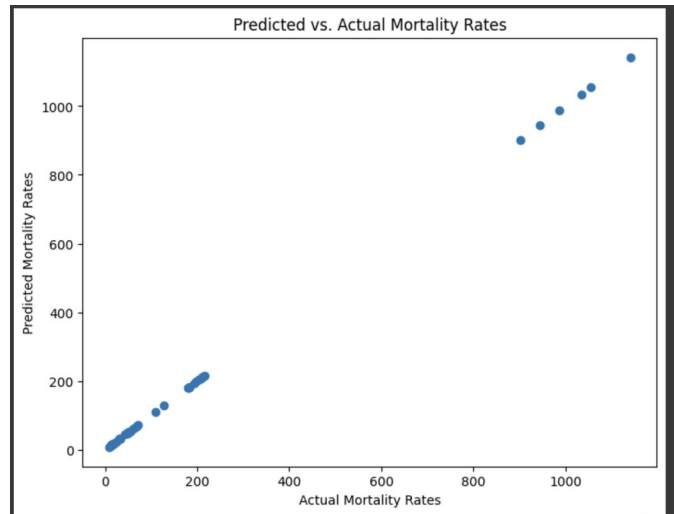


Figure 8.2.3 Visualisation of Predicted vs Actual Mortality Rates

(Source:Self-cretaed)

The researchers adopted to develop and create a Residual Plot display in order to examine mortality rate computations. This wonderful tool allows a fast assessment of model efficacy by comparing the actual and projected mortality rates. To aid in emphasising apparent biases or trends, the Residual Plot presented a visual picture of the gap between the actual and predicted values (Braiman., 2020). This procedure strengthened the researchers' capacity to analyse the accuracy of death rate projections and allowed for a thorough review of the model's virtues and limits, which improved mortality prediction techniques.

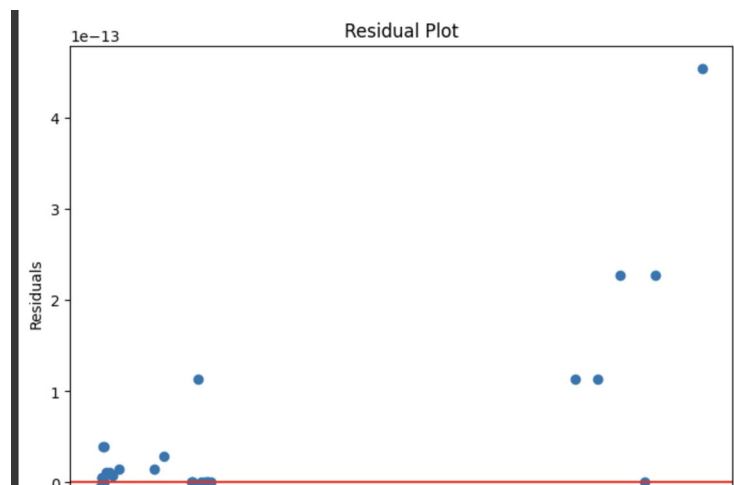


Figure 8.2.4 Residual Plot

(Source: Self-created)

8.3 Summary

The model exhibits proficiency in mortality rate prediction in the testing report. Its exceptional prediction accuracy implies a trustworthy ability to examine facts and generate forecasts. The model's strong performance indicates how beneficial it may be for establishing healthcare programmes and activities that lower mortality risks.

Chapter 9: Project Management

9.1 Schedule Development

Task Name
Mortality Rate Prediction
1.0 Initiation
1.1 Requirement Analysis
1.2 Project plan and scope development
2.0 Data Collection
2.1 Jounrla collection
2.2 Literature Review
2.3 Dataset collection
3.0 Model Developmentt
3.1 Algorithm choosing
3.2 Software Development

3.3 Software Evaluation
4.0 Analysis
4.1 Results and Findings
4.2 Discussion
5.0 Project Closure
5.1 Future Work
5.2 Documentation
5.3 Project Submission

Table: WBS

(Source: MS Project)

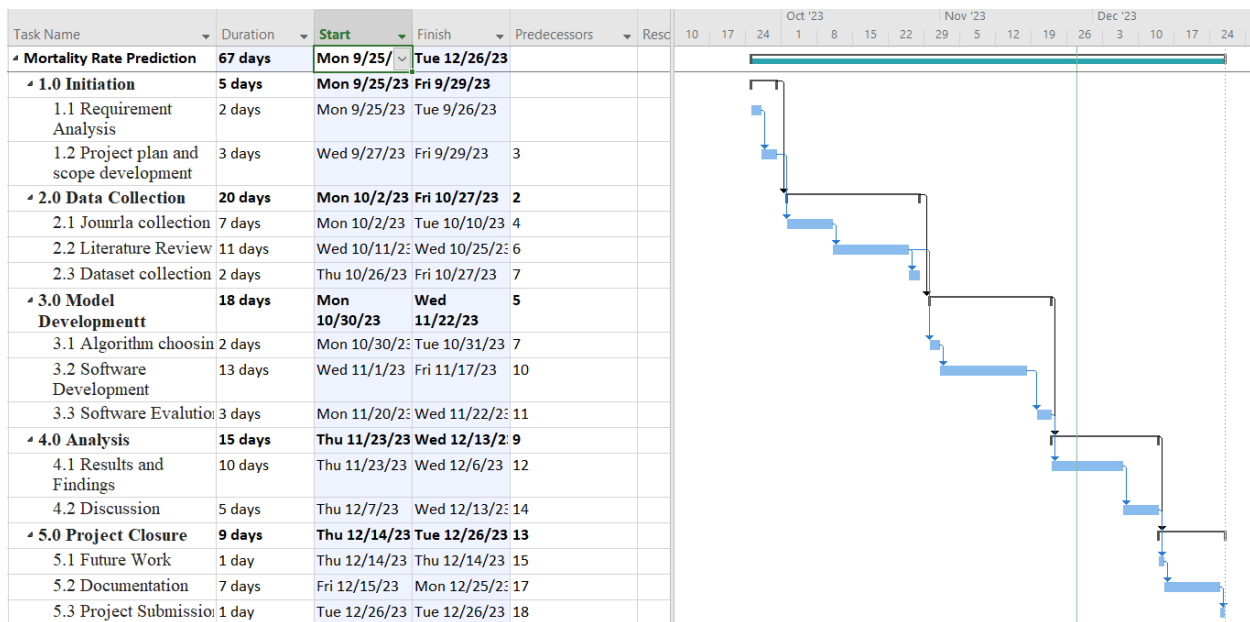


Figure 9.1: Gantt Chart

(Source: MS Project)

9.2 Risk Management

Risk	Probability	Impact	Action Plan
Scope Creep: Due to the time consumption of the project and its complexity, the chances of scope creep were there.	Moderate	High	The requirements of the project were evaluated after the completion of each chapter or activity. This is how it was ensured that any feedback or change in requirement would be effectively incorporated.
Time delay: The project has a specific delivery date from the university, and this is why time constraints were there.	Moderate	High	The Gantt chart was developed in such a way that any chances of schedule delay would be mitigated effectively. The resource levelling procedure was also used when needed. This signifies more time was given each day to mitigate any possibility of schedule delay as a priority.
Software performance: It is possible that the software would not fulfil all the requirements. The project's success is dependent on this software development process, and this is why it is a major risk.	Low	High	The requirements of the software were developed based on the aim and objectives. The target was to fulfil all of those requirements. One of the main things was to ensure the quality of the data which was collected already. The dataset was collected from a relief website mitigating the literature gap of this research process. This is how the risk of software

			performance was evaluated.
Technical Risk: It is possible that the hardware of the project may have technical issues within the project, and this could delay the project significantly.	Low	High	A backup computer was there with all the software installed to mitigate any technical risk. The target was also to ensure a permanent internet connection without any interruption which was also achieved in this project.

Table 7.1: Risk Management

(Source: Self-developed)

7.3 Quality Management

Standards adopted	Techniques	Evaluate Outcomes
Managing the quality of the software and code effectively.	Reviewing the progress after every step of the coding.	The outcome was a quality code that maintained the quality standards.
Fulfilling all the project objectives.	Review the project documentation with each project goal after the completion of each chapter.	The research process was able to fulfil all the objectives.
Maintaining professional standards of documentation.	Looking for structural mistakes, grammatical mistakes and so on every step of the documentation.	The project documentation was effectively done.

7.4 Professional, ethical, social, and legal Considerations

Professional Considerations

As a researcher it was important to fulfil all the professional considerations of the research process. This includes the guidelines, and structure provided by the university. The referencing style and time of submission were also followed to ensure that the code of conduct was followed. The systematic structure of a report was also followed which has been submitted to the university.

Ethical Considerations

The research topic is sensitive, and it is directly linked to public health. This is why it is essential to consider ethics and morality related to the topic.

1. While collecting the secondary dataset from the governmental website, the target was to ensure that no demographic information would be collected. This would ensure the privacy of the people, which is an important aspect of research ethics.
2. It was important that the research presented honest remarks to answer the research questions. As this could be useful for public health, the target was to reduce bias in the research. This is why positivist philosophy was used. This means the research has presented objective findings based on the scientific method (Armondet *al.*, 2021). Any personal remark has been made with justification so that bias from this research process can be removed.
3. The research paper has used a fair amount of secondary journals to establish the background of the research topic. The target was to ensure that all of those journals had been referenced properly so that the chances of plagiarism could be removed from this research process (Droletet *al.*, 2023).

These are the primary ethical and moral considerations that had to take place to complete the research process.

Social Considerations

While conducting this research process, it was ensured that it would not be harmful to the general public. The research problem is very much a social problem, and this will have a significant impact on the society. This is why no demographic information about people was used in the research process. Further, it was ensured that the research documents would not be breached before the completion. Then, it could have been used illegally. This is why the research document was stored in the cloud storage before its completion.

Legal Considerations

It was assessed that the data protection law was followed throughout this research process, as already mentioned. Any regulations related to medical field was not broken through this research process as well.

Chapter 10: Critical Appraisal

Positive Aspects

1. This research process has successfully developed software that can predict mortality graphs.
2. The project evaluation shows that the software has provided a lower error value.
3. The research has met all the objectives effectively

Negative Aspects

1. The research process does not use any primary data, and it is effective for medical research if primary data is used.
2. The project has not developed a fullscale working software for mortality rate detection, which could be done as the future scope.

Chapter 11: Conclusion

11.1 Conclusion

To conclude, Machine learning algorithms, one of the main applications has been in the healthcare sector. Mortality rate detection and prediction is one of those application. The research aims to develop a software model which would predict mortality rates based on machine learning. The governmental website catalog.gov was used to collect the dataset which consists of data of death from various causes. The linear regression algorithm was used to predict the mortality rate. The developed model performed well, and it showed a high accuracy rate for predicting the morality date due to different causes. It was found in recent years Covid-19 has been one of the primary reasons behind the death of people, which is aligned with the real-world findings.

11.2 Achievements

With respect to the objectives, the research has fulfilled all the objectives:

1. To collect and prepare a dataset using which mortality patterns can be understood was the first objective. The dataset from catalog.gov was collected and used to train and test the model to understand the morality patterns.

2. Identifying the best features which are contributing to the mortality of the population was the 2nd objective. The causes and features of death, including COVID-19, drugs, cancer and other diseases, were evaluated in this research process as well.
3. Developing the software model using ML algorithms for mortality rate prediction was the third objective. Using linear regression, the model was developed, and mortality patterns and predictions were evaluated effectively.
4. To perform different evaluation techniques to understand the success of this software development was the final objective. RMSE technique was used to evaluate the software model. It was evaluated that the developed model has performed effectively.

It can be said that this research process has fulfilled all the objectives and developed the model as per the requirements.

11.3 Future Work

In the future, advanced ML algorithms can be used effectively. This includes various deep learning models to understand the intricacies of different features. In the future, the time-series analysis and geographical data of the mortality rate can also be developed. This way, the mortality rate for a specific region for a specific cause can be developed along with making comparisons between different places. The mortality rate prediction for specific causes like cancer, covid-19, or smoking can specifically be assessed in future studies as well.

Chapter 12: Reflection

Using Gibbs's reflective cycle, the reflection will be done.

Description: As part of the research project, I have conducted research on mortality rate prediction using machine learning. I was successfully able to complete both the documentation and software processes.

Feelings: I am happy right now as I am almost at the end of completing the research. However, I was tense initially due to the complexity of the project. With time, I gained confidence in being able to complete the research process.

Evaluation: I learned a lot of things from this research. This includes how to conduct a research process, project management techniques, and knowledge about Python coding. The thing that did not go well was the time management. I could not complete some of the work on time and this is why I had to work more hours to prevent any delay.

Analysis: The possible reason behind the time delay happened due to my lack of experience in conducting research this big. However, I was able to mitigate the schedule risk effectively.

Conclusion: If I have the chance to complete the project again, then I would like to plan my schedule in more depth. This way, I would be able to gain more background knowledge about the research topic.

Action Plan: In future, I would like to improve my coding skills, which will be helpful for me in my professional life. I would also like to improve my project management skills, which are equally important along with technical knowledge.

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